

Classifiers

P81

9/30

2 goals:

① Ch4 wrap (accuracy metrics)

② Ch5, 2 topics

① Cross validation (test accuracy approx)

② Bootstrap for conf/SE estimates

Ch4 Classification Accuracy

Y in $\{Y_1, Y_2, \dots, Y_k\}$
 Predict $\hat{Y} \sim Y$ via (log reg)

Assess model accuracy? More complex than regression (R^2, MSE)

Spam email
 classif
 (spam/not)

Confusion matrix
 (k=2, 2 classes)

Pred (via model)

If data imbalanced, misleading.

	Spam (Pos)	Not (Neg)	Total
Spam (Pos)	40 (TP) TPR	10 (FP) TNR	50
Not (Neg)	20 (FN)	30 (TN)	50
Total	60	40	n=100

P = pos pred
 N = neg pred
 T = true (corr)
 F = false (incorr)

Accuracy: $Acc = \frac{TP+TN}{TP+FP+FN+TN} = \frac{40+30}{100} = 0.7$ (in $[0,1]$)

Focus instead on capturing true pos (TP)

AKA Sensitivity or recall

Specificity

Spam class acc.
 True pos rate

True pos: $TPR = \frac{TP}{TP+FN} = \frac{40}{40+20} = \frac{40}{60} = \frac{2}{3} = 0.6$ (in $[0,1]$)
 perc of pos class got right

True neg rate: $TNR = \frac{TN}{TN+FP} = \frac{30}{30+10} = \frac{3}{4} = 0.75$

Precision: $Prec = \frac{TP}{TP+FP} = \frac{40}{40+10} = \frac{4}{5} = 0.8$ (in $[0,1]$, perc classify spam correct)

issue spam misclassify (Precision-recall balance)

$$F_1\text{-score: } F_1 = \frac{2 \cdot \text{Prec} \cdot \text{Recall}}{\text{Prec} + \text{Recall}} = 2 \cdot \frac{0.8(0.6)}{0.8 + 0.6}$$

$$= 2 \cdot \frac{\frac{4}{5}(\frac{3}{5})}{\frac{4}{5} + \frac{3}{5}} = 2 \cdot \left(\frac{4.2}{12+10} \right) = 2 \cdot \frac{8}{22} = \frac{8}{11}$$

$$= 0.72 \text{ (in } [0,1], \text{ "avg" of } 0.8, 0.6)$$

Replace with "better" metric.
 Robust to imbalance classes.
 Prec = Recall = 1 (TPR)
 skewed by smaller

All of confusion matrix metrics done. ✓
 One more view: Holistic model assessment.

(Receiver
operating
Curve)

WWII

ROC curve:

(Recall) TPR 0.6

Summarize into a single score

AUC in $[0.5, 1]$

(Area
under
curve)

(Perf model)
sample
sep rate

ROC
Curve

AUC = 1 (best)
= 0.5 (worst)

No info rate

0.25 FPR
(1 - specificity)

Let vary $\rightarrow p=0.5$ (DB)

(Neg) Not spam prob $\rightarrow P(X=x|Y=0)$

$P(X=x|Y=1)$ spam prob (pos)

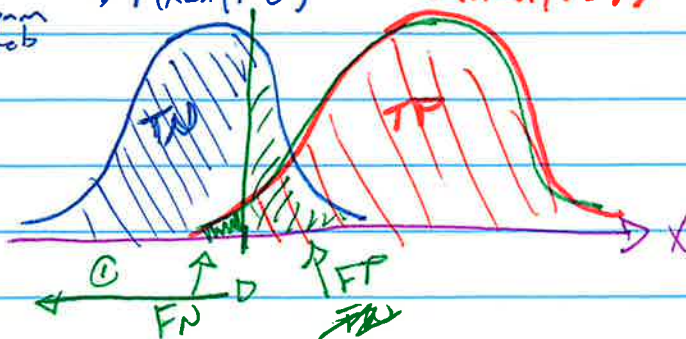
If decrease D

TN $\downarrow$ ☹️

TP $\uparrow$ 😊

FN $\downarrow$ ☹️

FP $\uparrow$ ☹️



Multiclass case: Y in $\{Y_1, Y_2, \dots, Y_K\}$, get all metrics as avg of many 1 vs rest metrics
K times

Ch 5 Resampling methods

Idea: Sample train data for

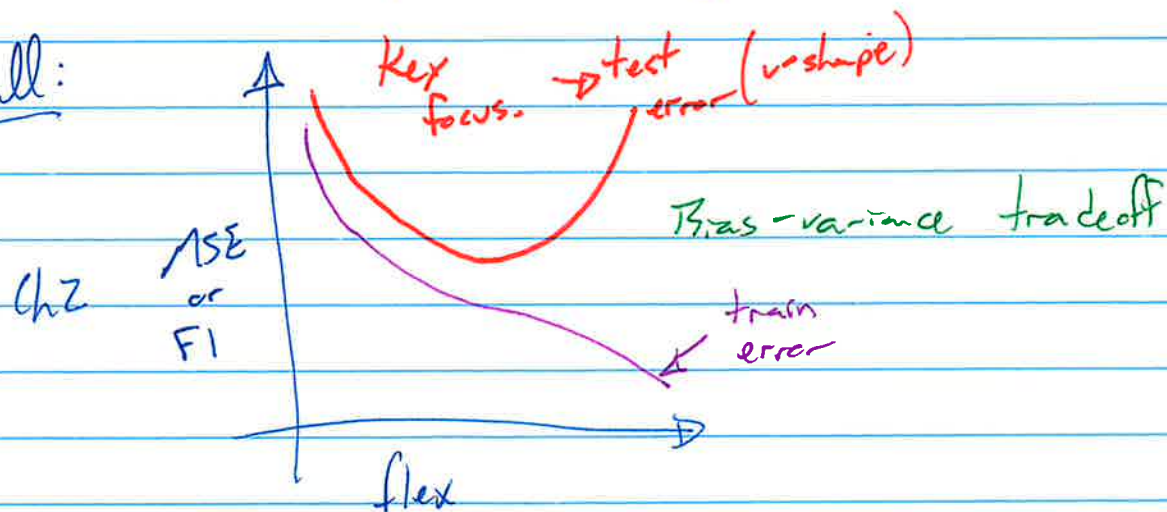
- ① Approx test accuracy
- ② Approx SE formulas

Cross-validation: Use train set to approx test accuracy

Issue: Usually don't have a test dataset $\frac{1}{c}$

- (i) Use all data as train to get test model
- (ii) If did keep test data, what is "best" choice? 😊

Recall:

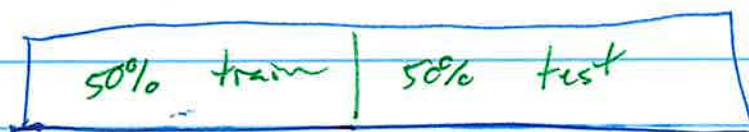


Use train dataset to approx test error. Many ways.

Cross validation

- ① Validation set
- ② Avg many validation sets
 - LOOCV
 - K-fold CV

Single validation set:



Random partition
once.

Entire train
data (x_i, y_i)

Issue: Using single partition sees variability in
both train/test error.

Soln: many CV partitions.

② model selection
& building.

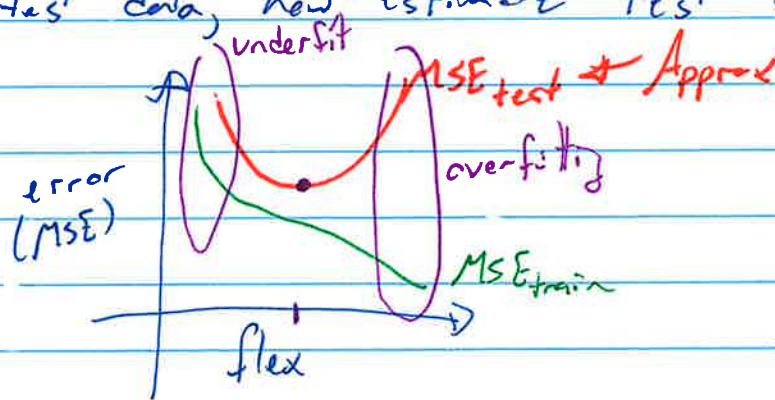
Ch5 Resampling methods

- ① CV (model assessment) 10-2
- ② Bootstrapping (model reliability) PG 1

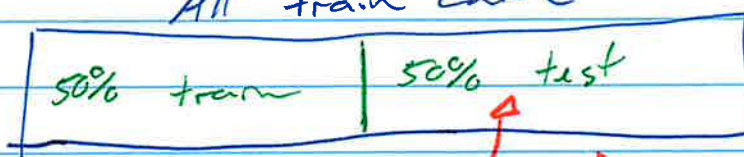
Cross validation

In practice, use whole sample for training model

- Gives best model
- No test data, how estimate test error



Strategy: Split train data into 2 parts



single validation split.

approx of MSE_{test}

Cons: ① Random single splits creates variance in

MSE_{test} estimate

② Single split ~~over~~ underestimates MSE_{test}

model sees little data.

2 fixes: ① LOOCV

② k-fold CV

sample size

Pg 2

Test

Train

n
CV
sets

1 2 3 ... n

Test

125

1	2	3	-	-	-	-	-	↗
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Train

;

trần

Test

Train

1	2	3	...	n-1	/
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the model
pred.



LOGCV
test error

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n MSE_i, \quad MSE_i = \frac{1}{1} (y_i - \hat{y}_i)^2$$

Test only

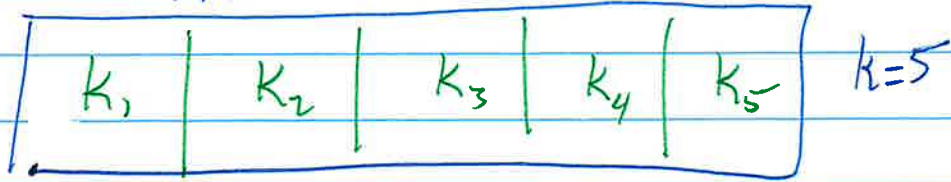
Pros ~~each~~ B/c each model uses most of sample,
get a closer estimate of MSE_{test}

Cons ① MSE; may vary highly
② n big, too many models (ineff)

Best practice ($K=5, 10$)
 k -fold cross validation

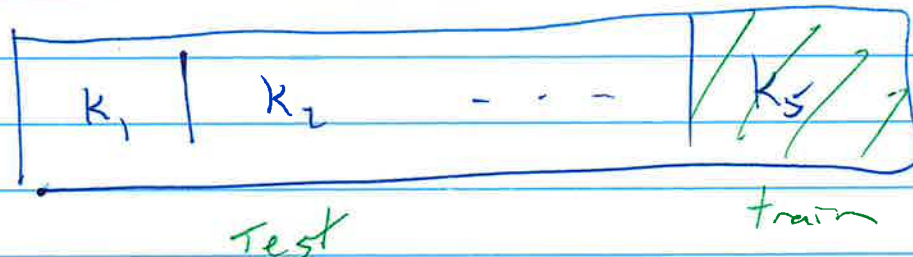
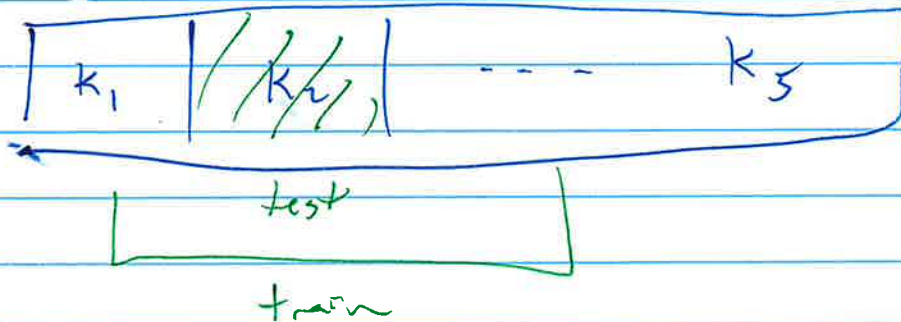
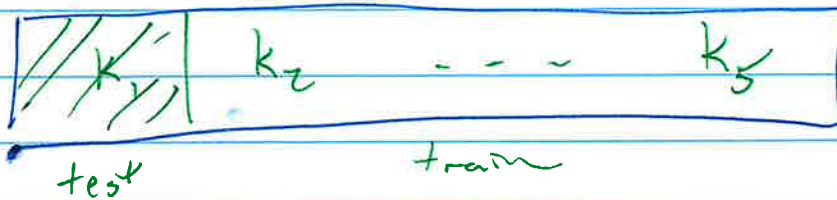
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All test data



k equal partitions

$\hookleftarrow$ CV sets



$$MSE_{test} \approx CV_{(K)} = \frac{1}{K} \sum_{i=1}^K MSE_i$$

$$MSE_i = \frac{1}{n_K} \sum_{i=1}^{n_K} (y_i - \hat{y}_i)^2$$

$\nwarrow$ k models

Prq/con table.

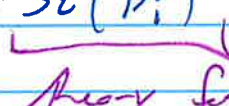
Bootstrapping (for model reliability)

Recall: $Y = f(X) + \epsilon = \beta_0 + \beta_1 X + \epsilon$

$$Y \approx \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X \quad (\hat{\beta}_i \text{'s via least squares})$$

Reliability: If train data changes slightly, how do $\hat{\beta}_i$'s change?

95% CI: $\hat{\beta}_i \pm 2 \cdot SE(\hat{\beta}_i)$


 Any formula (only for nice models)

Use bootstrapping to approx $SE(\hat{\beta}_i)$ for more complex models. 